

Assignment

Design and implement a complete deep learning solution for a real-world problem (such as image classification, biomedical image analysis, natural language processing, or time-series prediction) using an appropriate neural network architecture.

Your assignment must include the following components:

1. **Model Selection & Architecture**

- Justify the choice of network (MLP/CNN/RNN/LSTM/Pretrained model such as ResNet, VGG-16, DenseNet, EfficientNet or U-Net).
- Describe the architecture with layer-wise details, activation functions, and output formulation.

2. **Mathematical Explanation**

- Explain the forward pass and the backpropagation learning process.
- Show how the loss function and an optimizer (Gradient Descent/Adam/RMSProp etc.) update the weights.

3. **Regularization & Hyperparameter Tuning**

- Apply at least two techniques to address overfitting (e.g., Dropout, L1/L2, data augmentation, learning-rate scheduling).
- Analyze their impact on performance.

4. **Performance Evaluation**

- Train and test your model using a standard dataset.
- Present accuracy/loss curves and appropriate evaluation metrics.

5. **Critical Analysis**

- Discuss vanishing/exploding gradient (if applicable).
- Compare your model's performance with another existing architecture.

6. **Application Perspective**

- Explain how the developed system can be used in a real-life application domain.